

Training Configuration and Summary:

Dataset used: rssi_odom_dataset_12apr25_isolado.parquet

model_type: LSTM
group_sec: 200
timesteps_orig: 60
batch_size: 256
min_units: 16
max_units: 256
units_step: 16
dropout_min: 0.2
dropout_max: 0.3
dropout_step: 0.1
activation_functions: ['tanh', 'relu', 'elu']
optimizers: ['adam', 'rmsprop', 'Nadam']
min_layers: 1
max_layers: 2
max_epochs: 100
hyperband_iterations: 3
group_msec: 200
samples_per_sec: 5
forecast_horizon_sec: 4
forecast_steps: 20
timesteps: 60

LSTM Multi-Step Training Report

Report Generated: 2025-08-01 16:09:07

Training Start: 2025-08-01_16-03-42

Training End: 2025-08-01_16-08-17

Training Duration: 4.58 minutes

Predictions Start: 2025-08-01_16-08-17

Predictions End: 2025-08-01_16-08-30

Predictions Duration: 0.22 minutes

Best Hyperparameters:

Number of Layers: 2

Activation: tanh

Optimizer: Nadam

Dropout Rate: 0.20

MAE: 2.8384

MSE: 10.7527

RMSE: 3.2791

R2 Score: -0.5390

Multi-Step Forecast:

Forecast Steps: 20

MAE: 0.0794

MSE: 0.0088

RMSE: 0.0938

R2 Score: 0.4252

Dataset size before down-sample: 31,097 rows
Dataset size after down-sample: 1,870 rows

First 5 timestamps BEFORE:

1744464597710
1744464597720
1744464597744
1744464597757
1744464597783

...

Last 5 timestamps BEFORE:

1744464971366
1744464971377
1744464971390
1744464971402
1744464971414

First 5 timestamps AFTER:

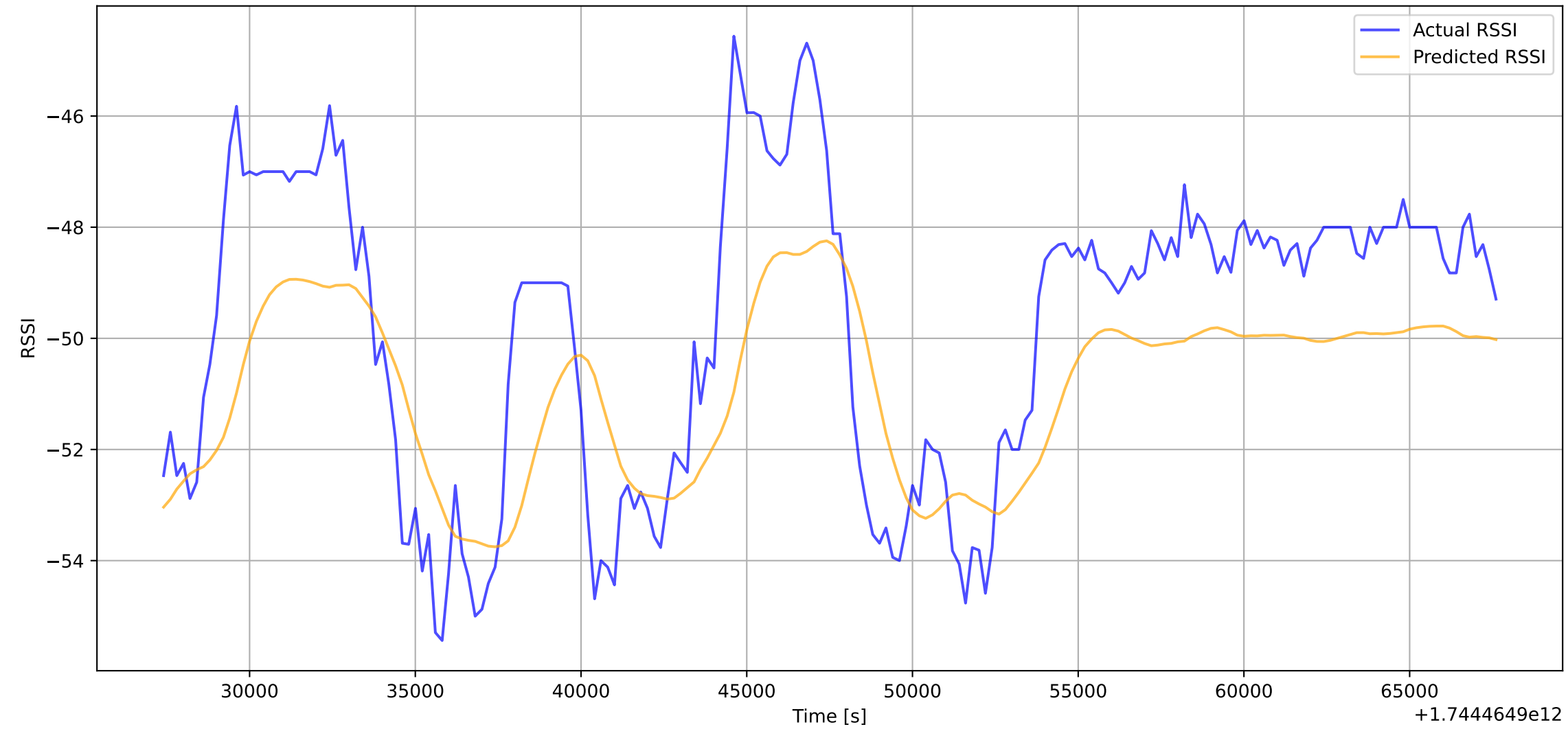
1744464597710
1744464597812
1744464598008
1744464598210
1744464598404

...

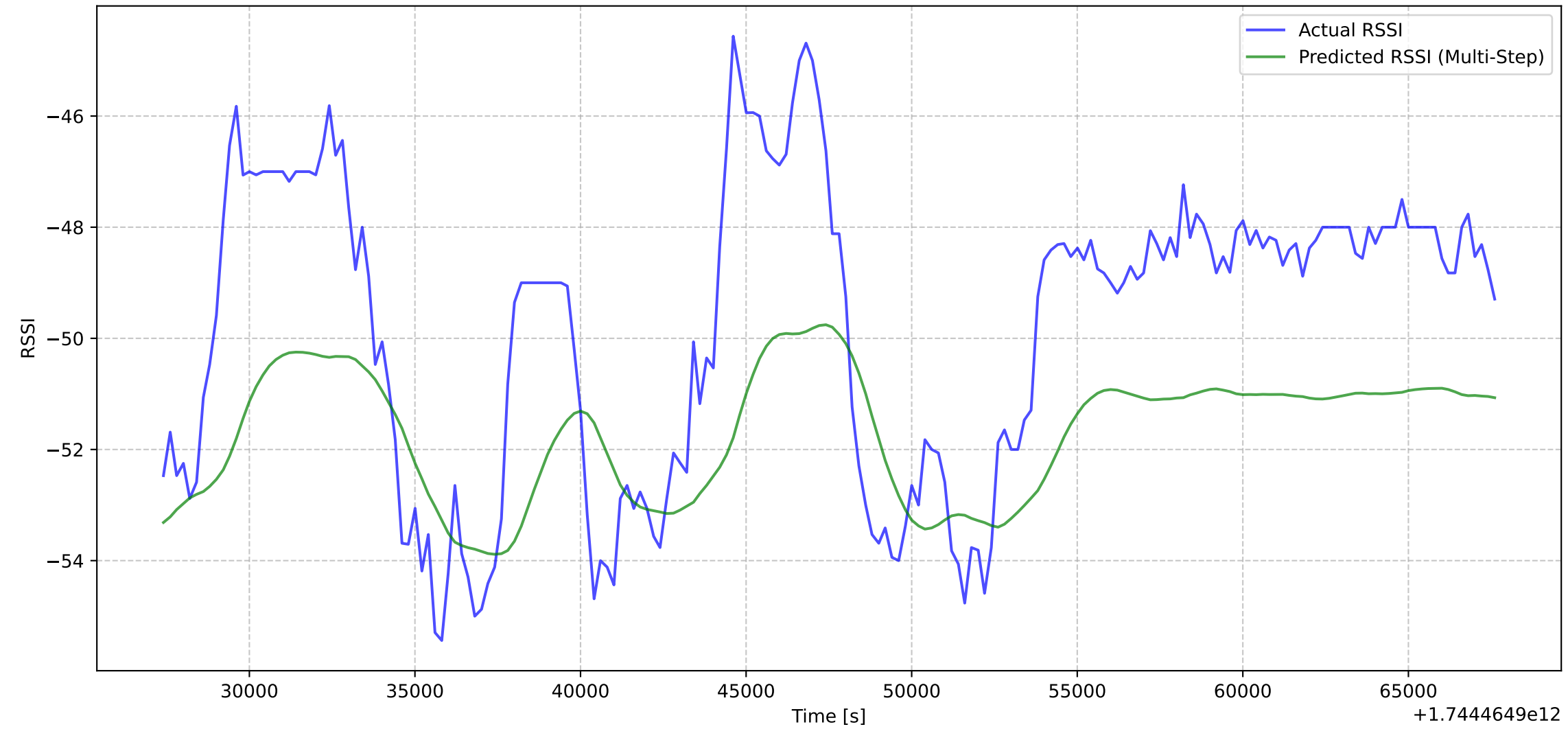
Last 5 timestamps AFTER:

1744464970609
1744464970802
1744464971004
1744464971209
1744464971402

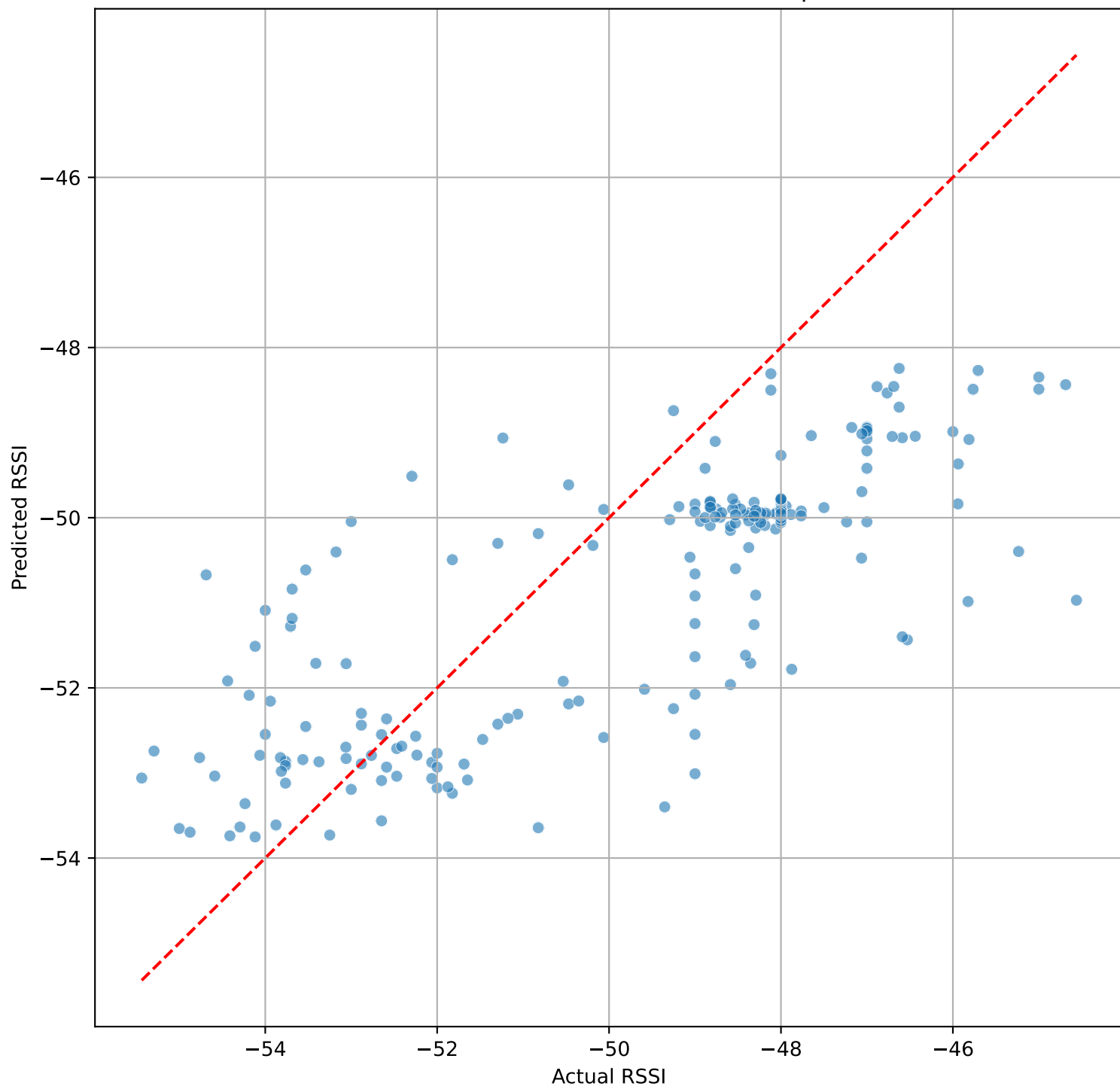
Actual vs Predicted RSSI (First Step)



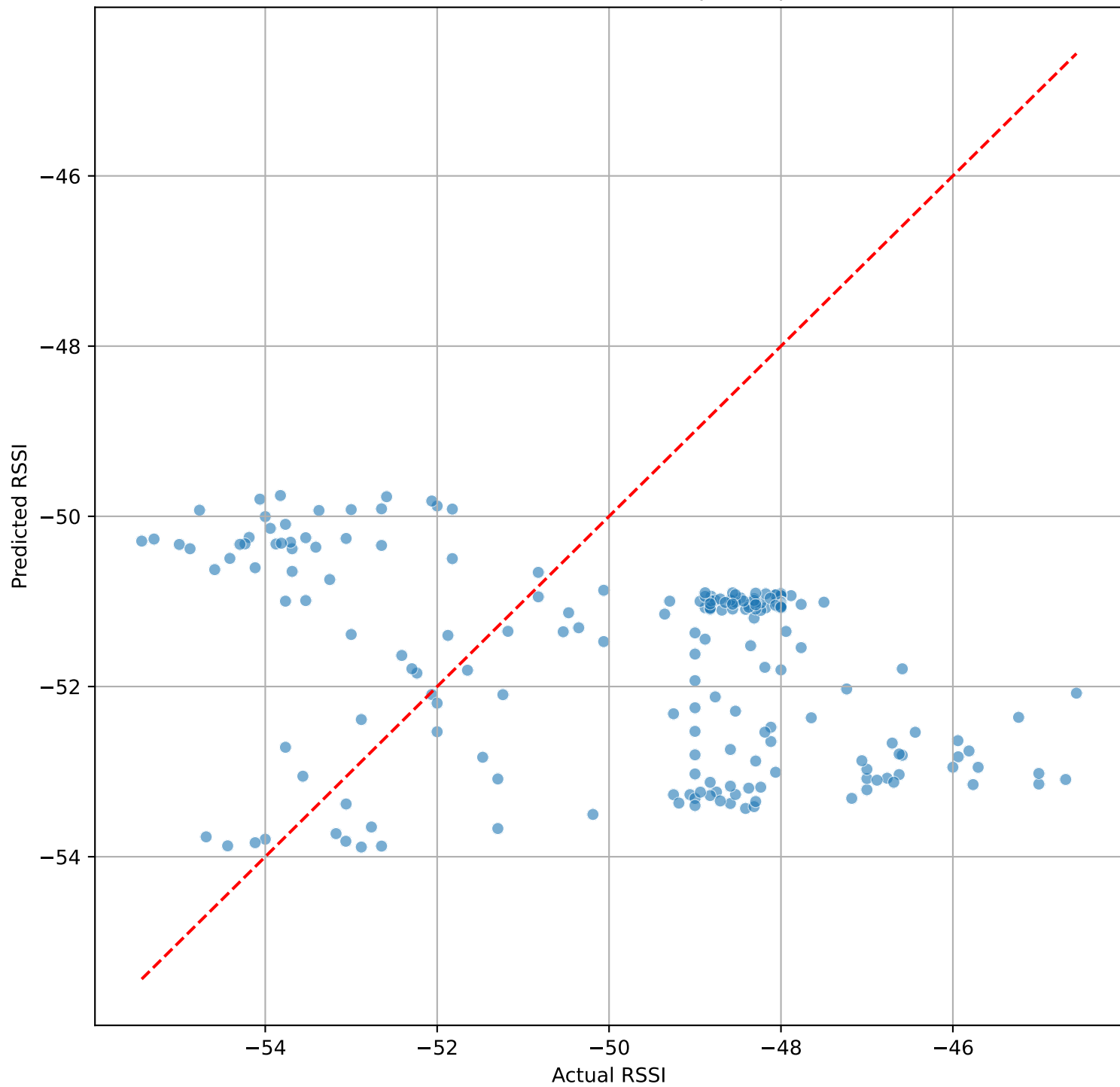
Actual vs Predicted RSSI (Multi-Step (step 20 $\approx$ 4.0 s))



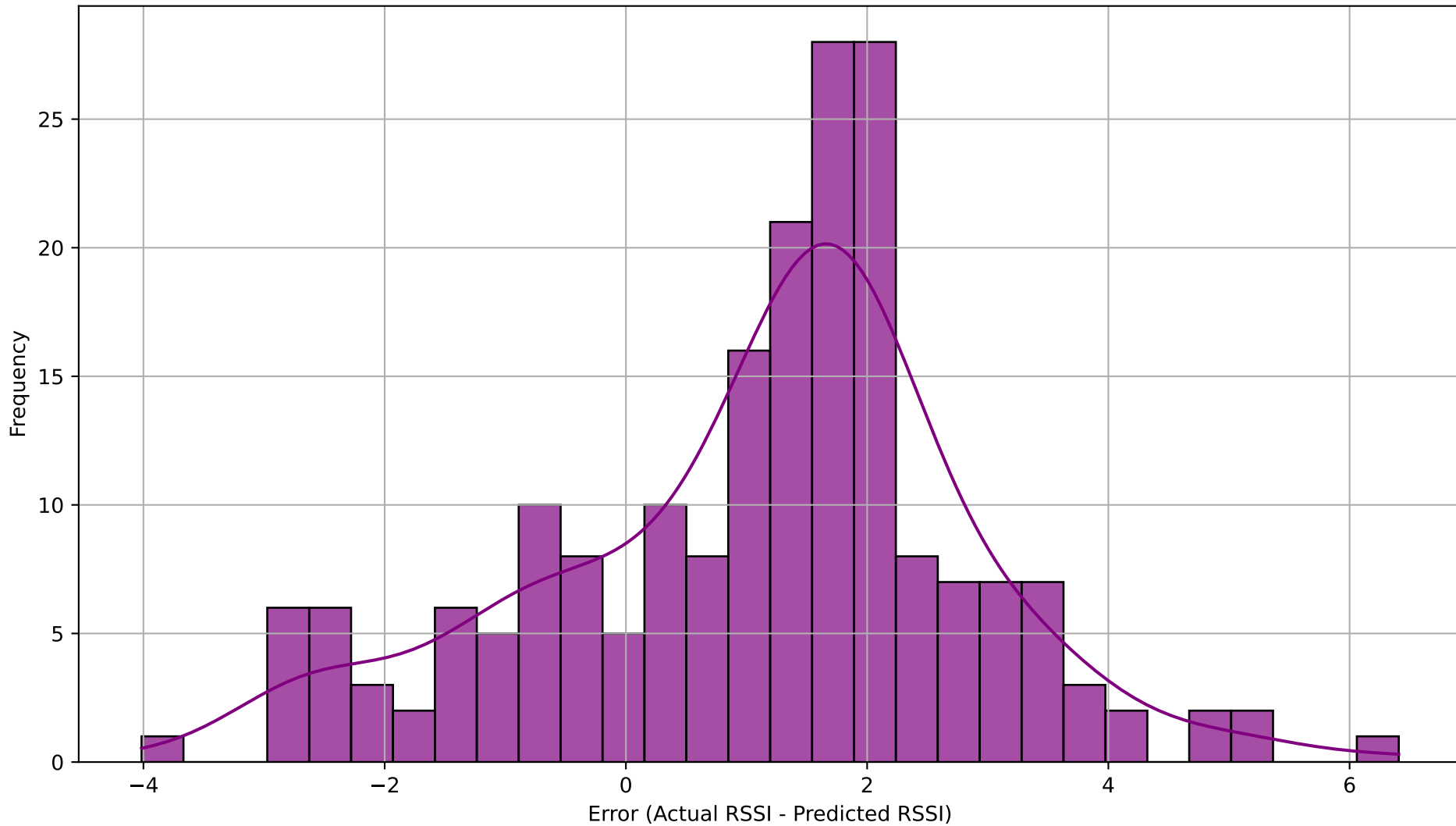
Actual vs Predicted RSSI (First Step)



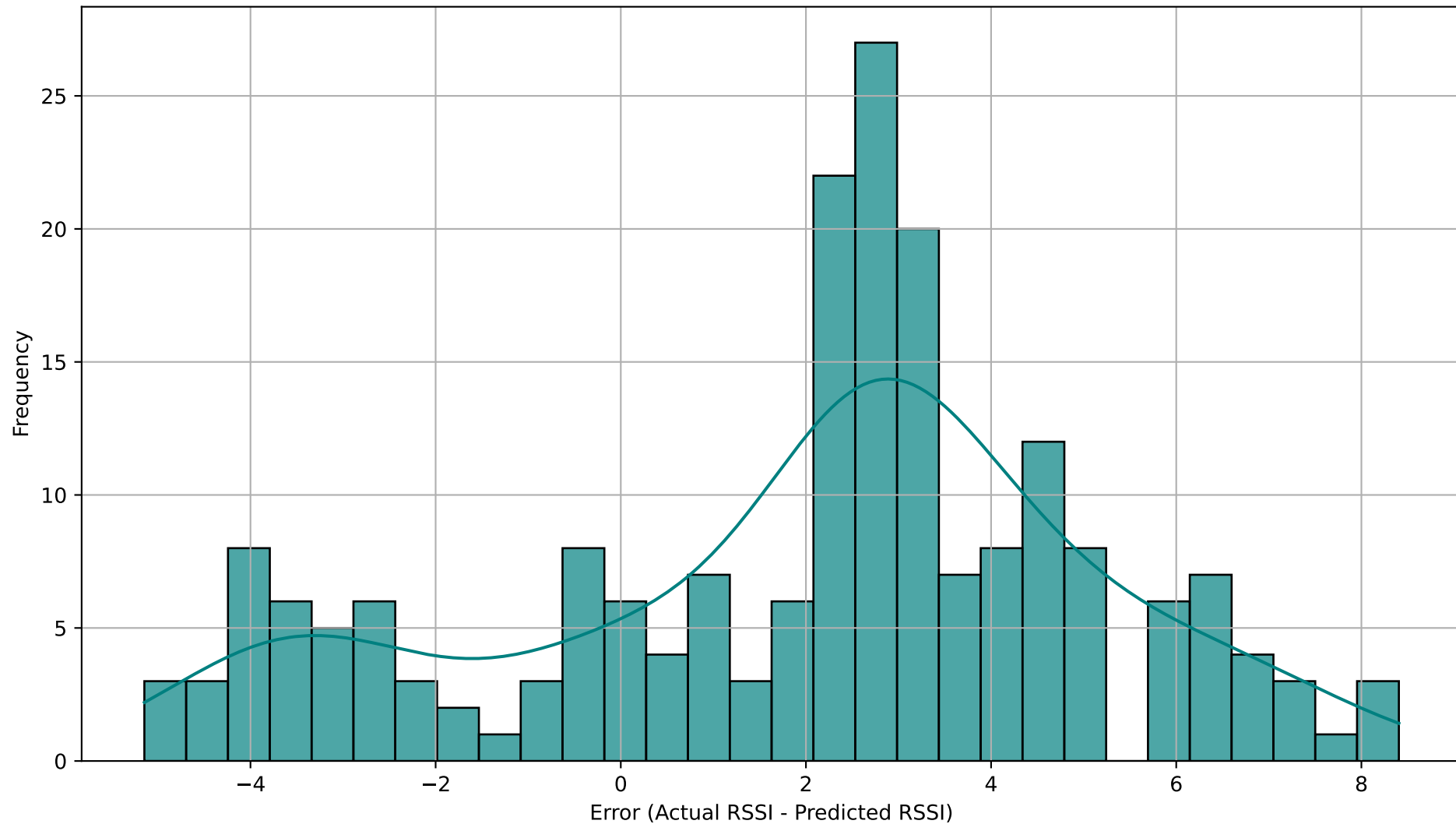
Actual vs Predicted RSSI (Last Step - step 20 $\approx$ 4.0 s)



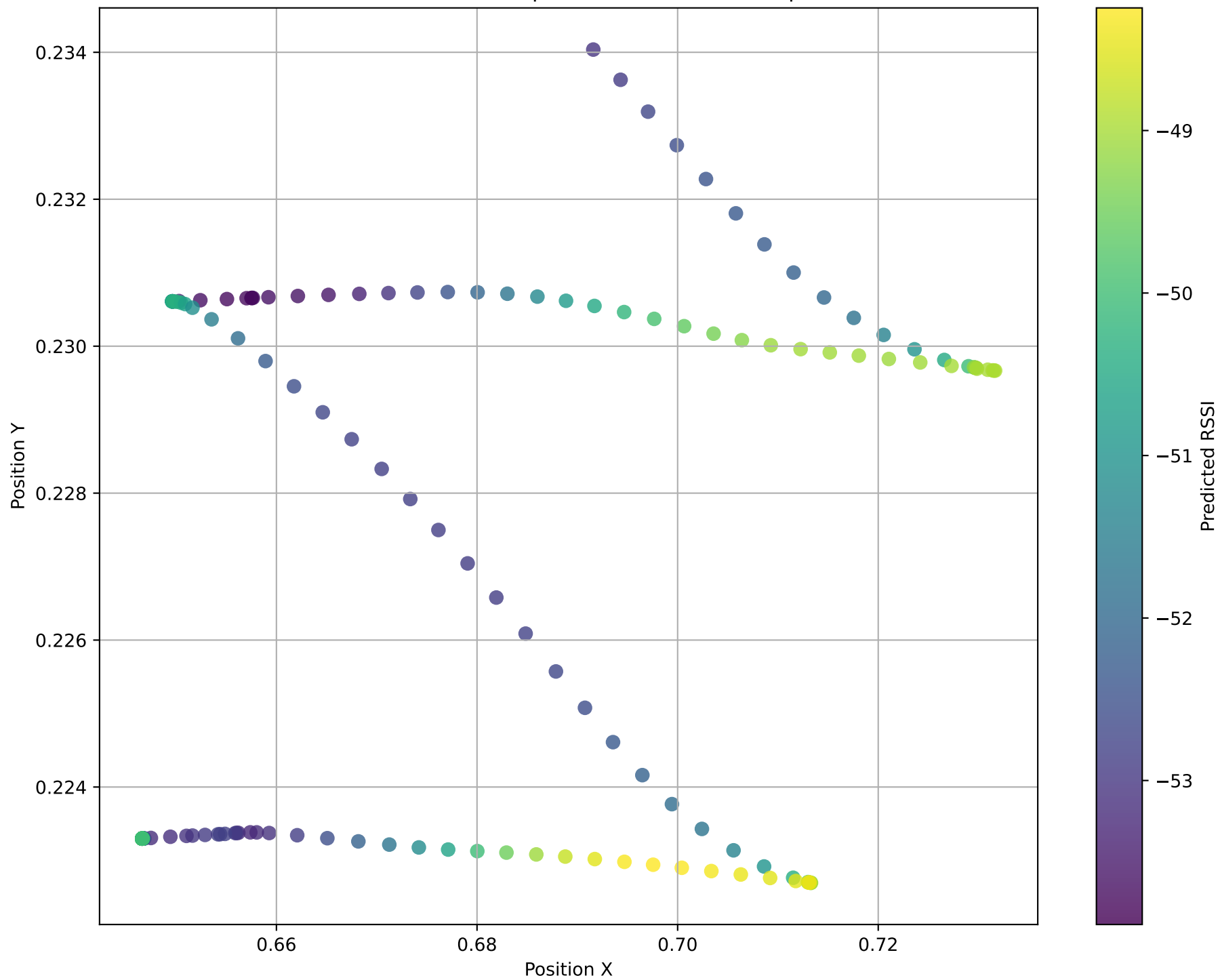
Distribution of Prediction Errors (First Step)



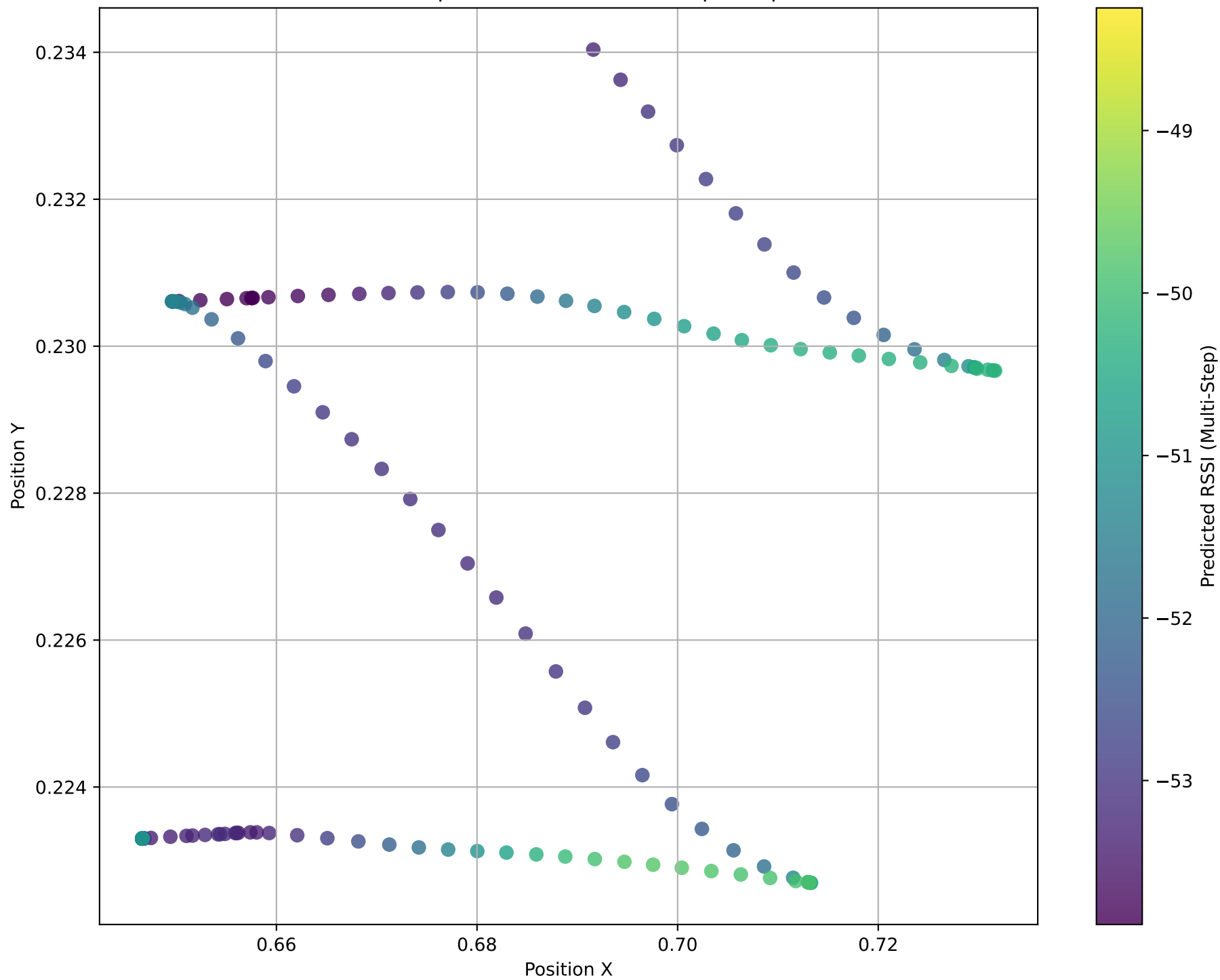
Distribution of Prediction Errors (Last Step - step 20 $\approx$ 4.0 s)



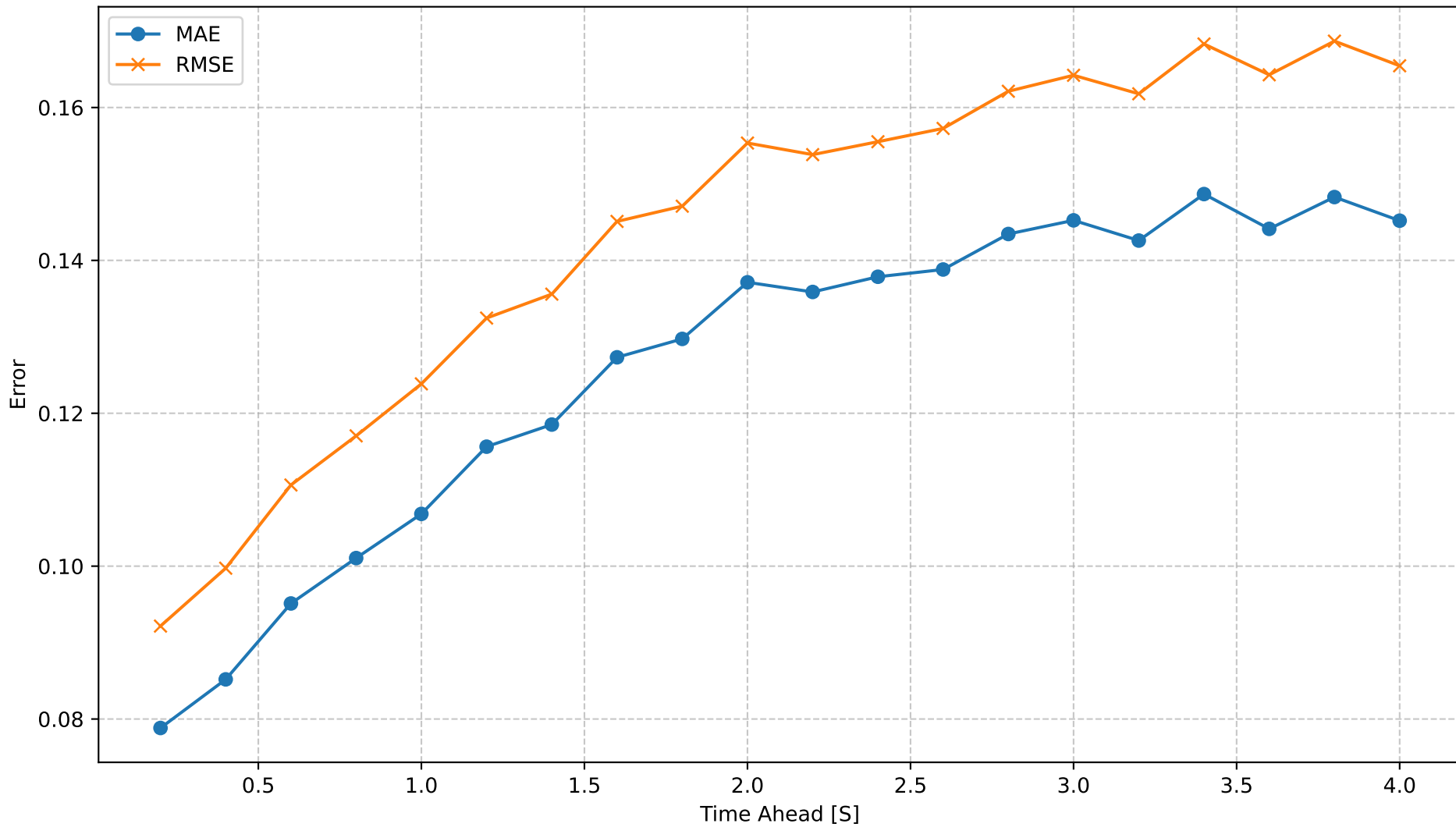
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 20 $\approx$ 4.0 s)



Multi-Step Forecast Error by Time Ahead



Resumo Automático

Dataset: rssi_odom_dataset_12apr25_isolado.parquet
Modelo: LSTM
Forecast steps: 20

MAE / RMSE (dB)

- 1º passo : 0.08 / 0.09
- Mediana : 0.14 / 0.15
- 20º passo : 0.15 / 0.17

Crescimento do erro: 84.2% → Degrada Rápido (> 50 %)

3 piores passos (MAE): 17 (0.15), 19 (0.15), 15 (0.15)

Recomendação:

Considere aumentar o histórico de entrada ou usar um modelo mais profundo para melhorar multi-step.